Viability Assessment of Real-Time AI Computer Vision for Nationwide Polling Station Monitoring: 2026 Armenian Parliamentary Elections

The integration of artificial intelligence (AI) and computer vision into electoral administration represents a profound paradigm shift in safeguarding democratic integrity. As the Republic of Armenia approaches the scheduled 2026 Parliamentary Elections, the political environment remains highly polarized, having been shaped by recent geopolitical conflicts, municipal amalgamations, and intense domestic competition.[1] Historically, the monitoring of the approximately 1800 to 2000 polling stations across Armenia has relied heavily on manual observation by domestic civil society organizations, international bodies such as the OSCE Office for Democratic Institutions and Human Rights (ODIHR), and political proxies.[5] While effective in isolated contexts and essential for democratic legitimacy, human observation is inherently constrained by physical fatigue, limited fields of view, logistical expenses, and the fundamental impossibility of maintaining ubiquitous, uninterrupted coverage across every precinct simultaneously.

The proposition to develop and deploy a real-time AI computer vision system to monitor live video streams from approximately 1800 polling stations simultaneously introduces an unprecedented scale of automated surveillance designed specifically to detect electoral fraud and procedural violations. This report provides an exhaustive, multi-disciplinary analysis of the viability of this proposed system. By synthesizing global precedents in AI-driven public surveillance, establishing rigorous alignment with the specific legal thresholds of the Electoral Code of the Republic of Armenia, analyzing extreme technical architectural constraints, and mapping the operational synergies between artificial intelligence and human oversight, this document evaluates whether such a system is practically feasible for a nationwide deployment.

1. Global Market & Academic Precedents in AI Election Monitoring

The application of computer vision and machine learning within public administration and electoral environments is an emerging, yet rapidly accelerating, discipline. While a nationwide, real-time AI monitoring apparatus designed explicitly for comprehensive fraud detection across all polling stations remains exceptionally rare, several regional implementations and rigorous academic proofs-of-concept provide a foundational blueprint for the proposed Armenian system. Understanding these global precedents is critical for anticipating both the capabilities and the inherent vulnerabilities of algorithmic surveillance in highly sensitive political contexts.

1.1 Real-World Implementations: The Kerala Queue Monitoring and VMukti Paradigms

One of the most prominent practical applications of artificial intelligence in an electoral context occurred during the assembly polls in Kerala, India. Driven by judicial pressure from the Kerala High Court, which mandated basic amenities and operational efficiency to prevent long delays that were characterized as "the death knell of democracy," the State Election Commission implemented a massive shift from standard laptop webcams to networked Pan-Tilt-Zoom (PTZ) cameras across more than 30,000 polling stations.[7] This deployment represents one of the largest integrations of public surveillance deep inside electoral logistics.[7]

The system in Kerala was designed primarily to perform real-time queue monitoring. It utilized edge-ready IP cameras that fed into local Network Video Recorders (NVRs), which then transmitted data to cloud-based analytical dashboards.[7] The underlying computer vision algorithms focused on analyzing crowd density, loitering patterns, and throughput rates, subsequently calculating estimated wait times and pushing automated, real-time alerts to a centralized panel.[7] Each alert included critical metadata such as the specific booth ID, a precise timestamp, and the estimated duration of the queue delay, allowing rapid response teams to dispatch additional staff or physical resources to congested stations.[7] While this specific implementation was focused on logistical optimization rather than direct fraud detection, the Kerala deployment established the fundamental viability of processing massive volumes of electoral video data. It demonstrated that hybrid edge-cloud inference could effectively slash bandwidth requirements and latency, resulting in early field reports indicating reduced voter abandonment rates.[7]

In the commercial and enterprise sector, surveillance vendors such as VMukti have developed highly specialized video management systems (VMS) tailored explicitly for election and large-scale public monitoring.[8] These enterprise solutions incorporate a suite of intelligent video analytics powered by deep learning models, including automated number plate recognition (ANPR), facial recognition, and sophisticated multi-site motion pattern analysis.[8] VMukti's systems represent a shift away from traditional, reactive surveillance—where video is only reviewed post-incident—toward proactive, real-time intelligence.[11] The algorithms are trained to identify specific behaviors, such as anomalous loitering, sudden chaotic motion, and directional flow changes in sensitive zones, executing these tasks without human intervention.[11] By continuously evaluating video frames and converting raw visual data into structured, searchable intelligence, these commercial systems prove that the hardware topologies and cloud architectures required for managing thousands of simultaneous streams are commercially mature and readily available.[9] However, transitioning these systems from detecting general "abnormal behavior" to identifying highly specific, legally defined electoral violations requires bespoke algorithmic tuning and rigorous constraint management.[11]

1.2 Academic Foundations and the Efficacy of Computer Vision in Governance

Beyond commercial deployments, the academic literature increasingly highlights the efficacy of AI-driven election monitoring systems to enhance transparency, minimize human error, and detect coordinated manipulation. Contemporary research has explored the creation of integrated, AI-driven digital platforms that synthesize electronic voting databases, natural language processing for social media analysis, and live monitoring feeds to identify abnormalities and potential risk factors in real-time.[13] In these proposed architectures, front-end interfaces developed in frameworks like React.js interface with backend Node.js and PostgreSQL databases, creating centralized command centers capable of rapid decision-making during active elections.[13] The findings from these academic models consistently confirm that automated surveillance significantly enhances response times and accountability in democratic processes, overcoming the inherent limitations of conventional human monitors who can only survey a fraction of voting sites.[13]

Furthermore, machine learning methodologies are increasingly being tested to identify polling places statistically at risk of election fraud. By analyzing historical voting data alongside real-time inputs, researchers have developed frameworks capable of estimating the extent of potential electoral manipulation, providing electoral management bodies with predictive threat assessments.[14] Similarly, in the broader realm of crowd analysis within public urban spaces—which serves as a direct analogue to the spatial dynamics of a polling station—computer vision algorithms are routinely tasked with identifying mass gathering events, tracking individual pedestrian paths, and classifying behaviors such as loitering or physical altercations.[15] Modern deep learning approaches in this domain treat abnormal crowd behavior as an outlier detection problem, utilizing complex spatio-temporal feature extraction to flag deviations from standard operational baselines, such as sudden shifts in walking speeds or unusual group formations.[16]

1.3 Algorithmic Vulnerabilities, Obscurity, and Primary Points of Failure

Despite the technological maturity demonstrated in Kerala and by enterprise vendors, global precedents highlight several critical points of failure and sociopolitical vulnerabilities inherent to the deployment of computer vision in uncontrolled public environments. A comprehensive review of AI implementation in Southeast Asian democracies (including Indonesia, Thailand, and the Philippines) revealed sharp postulations regarding algorithmic bias, a lack of data security, and the potential undermining of democratic values.[18] While systems aided by AI amplified administrative coordination, they simultaneously created a profound sense of public disquiet due to a high level of algorithmic transparency obscurity—specifically, the inability to explain unexplained anomalies in the data to the electorate.[18]

From a purely technical perspective, the primary failure points of real-time computer vision in polling stations include:

- **Algorithmic Bias and Demographic Disparities:** Machine learning models, particularly those involving facial recognition or pedestrian tracking, frequently exhibit varied accuracy rates across different demographic groups, raising severe ethical, human

rights, and constitutional concerns regarding surveillance bias and discrimination.[18]

- **Occlusion in High-Density Environments:** The absolute primary physical failure point in crowd analytics is occlusion.[20] In densely packed polling stations, the camera's line-of-sight to the voting booth, the registration desk, or the ballot box is frequently obstructed by voters, proxies, and commission members. This obstruction causes bounding box degradation, identity switching in tracking algorithms, and a complete failure of temporal persistence.[20]
- **False Positive Saturation and Alert Fatigue:** Surveillance systems optimized for high recall often generate an unmanageable volume of false positive alerts. For example, anomaly detection models may flag a voter taking an unusually long time in the booth due to a physical disability as engaging in "suspicious behavior," or they may flag a legally permitted proxy as an unauthorized loiterer.[15] If the false positive rate exceeds the review capacity of the human rapid-response team, the system suffers from alert fatigue, leading to operators ignoring genuine violations and causing a complete operational collapse.

2. Alignment with the Electoral Code of the Republic of Armenia

The overarching viability of an AI monitoring system hinges entirely on its ability to accurately and legally translate the abstract legal text of the Electoral Code of the Republic of Armenia into definitive mathematical rulesets that computer vision models can execute. The system must operate strictly within the boundaries of constitutional rights, particularly regarding voter privacy, free suffrage, and data protection, while simultaneously enforcing the rigid procedural mechanics of the voting process.

2.1 Translating Electoral Law into Computer Vision Tasks

To achieve actionable procedural oversight, specific computer vision architectures must be explicitly mapped to the corresponding legal violations defined by Armenian electoral law. The technical mechanisms required to detect these violations must be highly specialized, moving far beyond generic "security monitoring" to precision algorithmic compliance. Table 1 outlines this theoretical mapping between the legal framework and the necessary AI capabilities.

Article / Legal Requirement	Violation Description	Required Computer Vision Capability	Algorithmic Approach
Article 5: Secrecy of the Vote [24]	Unauthorized viewing of the voter's ballot, peering into the voting booth, or coercion inside the booth.	Pose Estimation, Head-Pose (Gaze) Estimation, 3D Spatial Geometry.	Mapping human skeleton joints (e.g., using YOLO-Pose) to detect if a second individual's posture or gaze vector crosses the defined 3D

			bounding box of the voting booth while occupied.
Article 65: Voter Assistance [25]	A single individual assisting more than one voter who is unable to complete the ballot independently throughout the day.	Person Re-Identification (ReID), Multi-Object Tracking (MOT), Embedding Extraction.	Extracting complex appearance embeddings of assistants and cross-referencing them continuously against an intra-day, localized vector database to flag duplicate spatial appearances.
Article 68: Summarization of Results [26]	Interference with the ballot box, or unauthorized persons touching ballots during the critical recount process at 20:00.	Object Detection, Hand-to-Object Interaction mapping, Spatio-Temporal Action Localization.	Tracking the ballot_box and ballot_paper classes, identifying instances where individuals not classified (via ReID) as official commission members physically interact with the materials.
Crowding / Capacity Rules [27]	Presence of unauthorized persons, excessive loitering inside the polling station, or violation of 50-meter perimeter rules.	Distance Measurement, Crowd Density Estimation, Dwell-Time Analytics.	Utilizing advanced tracking to calculate the dwell time of each tracked ID; triggering an alert if a non-official ID remains in the active zone beyond a legally defined temporal threshold.

2.2 Article 5 and Secrecy of the Vote: Boundary Conditions and Privacy by Design

Article 5 of the Armenian Electoral Code dictates that voting shall be held by secret ballot, emphasizing that secrecy of the vote is not merely a right, but a legal obligation of the voter, and that any control over the free expression of a voter's will is strictly prohibited.[24] This constitutional mandate poses a profound paradox for computer vision systems: the AI must rigorously monitor the voting booth to prevent coercion, intimidation, or vote-buying supervision, yet it must be absolutely physically and algorithmically prevented from capturing

the contents of the ballot paper itself.
To comply with this rigid mandate, the physical camera geometry and the software architecture must be standardized under the paradigm of "Privacy by Design." Cameras must be mounted at specific angles to capture the perimeter, approach, and entrance of the voting booths, but never the writing surface or the voter's hands while marking the ballot.
Furthermore, the AI architecture must enforce this privacy mathematically. A rigid, polygonal exclusion mask must be drawn over the interior volume of the voting booth in the video stream during the system's initialization phase; the AI must be hard-coded to ignore, drop, or pixelate all pixel data within this defined zone before any inference occurs.
Instead of looking inside the booth, the model relies on advanced Pose Estimation (utilizing architectures like YOLOv8-Pose) to analyze the vectors of human skeletons *outside* the mask. If the AI detects a primary skeleton inside the booth, it monitors the surrounding area for a secondary skeleton. If the coordinate vectors of the secondary person's neck, shoulders, and eye-line indicate they are leaning into or staring directly into the defined 3D volume of the booth, the system calculates the geometric intersection and flags a potential violation of voting secrecy and illegal coercion.

2.3 Article 65: Detecting Improper Voter Assistance via Person Re-Identification

Under Article 65 of the Electoral Code, a voter who is physically unable to complete the ballot papers independently may be assisted by another person, provided that the assisting person is not a registered proxy.[25] Crucially, the Code explicitly stipulates that the assisting individual is limited to providing assistance to only *one* voter, and the name of the person providing assistance must be formally entered into the registration book of the Precinct Electoral Commission (PEC).[25] Historically in post-Soviet electoral environments, the mechanism of "multiple assistance" has been widely exploited as a loophole for organized vote-buying and voter coercion, where a single operative repeatedly accompanies multiple voters into the booth throughout the day to ensure they vote for a specific candidate.
Detecting this systemic violation requires highly sophisticated Person Re-Identification (ReID) models. Standard object detection, which merely classifies a collection of pixels as "a person," is entirely insufficient for this task. When an individual is detected entering the proximity of the voting booth alongside a voter, the system must capture a localized, high-resolution crop of that assistant. The ReID model—typically a Vision Transformer-based architecture—processes this crop and converts it into a high-dimensional feature vector, known as an embedding.[28] This embedding serves as a mathematical signature of the individual based on clothing textures, color histograms, body proportions, and gait kinematics, completely independent of facial features.[28]
This embedding is securely stored in a localized, temporary vector database specific to that precinct. If a new embedding is captured later in the day assisting another voter, the system calculates the cosine similarity between the current embedding and all historical embeddings in the database. If the similarity score crosses a strict confidence threshold, indicating a highly probable match, the system triggers a "Multiple Assistance Violation" alert, allowing legal teams

to immediately verify if the PEC registration book reflects this illegal repetition.

2.4 Legal Constraints: Navigating the Armenian Law on Protection of Personal Data

The implementation of such advanced, pervasive surveillance must carefully navigate the rigid framework of the Armenian Law on Protection of Personal Data, which heavily regulates the collection, processing, and outflow of personal information, including biometric data.[29] Recent legislative amendments set to take effect in August 2025, which grant the Ministry of Internal Affairs 24/7 access to public surveillance cameras equipped with facial recognition, have sparked significant backlash and severe criticism from human rights organizations.[31] Civil society groups argue that the lack of clear legal safeguards, public oversight, and limitations on tracking individuals' movements violates the right to privacy and freedom of peaceful assembly enshrined in the Constitution.[31] Furthermore, Armenian data protection law dictates strict rules around automated decision-making and mandates immediate public announcements and notifications to the police in the event of an electronic data breach.[33]

To ensure absolute legality and maintain the vital trust of the electorate and international observers (such as the OSCE/ODIHR), the proposed AI system must absolutely avoid the deployment of facial recognition technology. Instead, it must rely exclusively on *Appearance-Based Person ReID*, which analyzes superficial clothing patterns and structural kinematics without identifying the actual human face or linking the data to national identity databases.[28] Furthermore, conforming to the restrictions surrounding automated decision-making [35], the system must be architected strictly as an *alert-only* mechanism; it cannot make automated legal determinations that invalidate a ballot or a precinct's results. Finally, to comply with data minimization and retention principles, all appearance embeddings and vector databases must be engineered with cryptographic self-destruction protocols, ensuring all biometric proxies are permanently deleted immediately following the legal certification of the election results.[33]

3. Technical Architecture & Constraints for 1800 Live Streams

Processing 1800 simultaneous video streams in real-time requires an enterprise-grade, distributed computing architecture capable of handling immense data payloads with sub-second latency. During the 2021 Armenian parliamentary elections, local telecommunications providers, notably Ucom, successfully deployed 1517 cameras to provide simultaneous live streaming of the voting and summarization processes over their fixed broadband and mobile networks.[38] This massive logistical effort, which required deploying over 200 fiber optic specialists to ensure network stability, serves as a crucial historical precedent, confirming that the national network infrastructure possesses the requisite backbone to sustain the required ingress bandwidth for a nationwide camera deployment.[3]

3.1 Edge-to-Cloud Topology and Network Ingestion

Assuming the deployment of 1800 RTSP (Real-Time Streaming Protocol) streams operating at a conservative resolution of 720p (to effectively balance network load against the pixel density required for accurate inference) and 15 frames per second (FPS), encoded via the highly efficient H.265 codec, each individual stream will require approximately 1.5 Mbps of sustained upload bandwidth. At a national scale, this equates to a total network ingress of roughly 2.7 Gbps terminating at the central processing facility.

To manage this throughput, a hybrid Edge-to-Cloud topology is optimal. Polling stations across the country will utilize fixed IP cameras connected to local routers provided by domestic ISPs (such as Ucom or Team Telecom Armenia).[3] To handle the massive, continuous ingestion without packet loss, protocol degradation, or buffer bloat, the central cloud architecture must employ a scalable, load-balanced media server cluster (utilizing technologies such as Kurento or a distributed NGINX RTMP/RTSP farm). This cluster will act as a robust reverse proxy, stabilizing the streams before feeding them into the GPU-accelerated AI inference pipelines.

3.2 Proposing an Optimal Tech Stack: The NVIDIA DeepStream Framework

For the real-time, parallel processing of thousands of concurrent video streams, the NVIDIA DeepStream SDK stands as the undisputed industry standard.[41] DeepStream utilizes a highly optimized GStreamer-based architecture that leverages dedicated hardware decoders (NVDEC) situated on NVIDIA GPUs.[41] This allows the system to decode H.264/H.265 streams entirely in hardware without burdening the general CUDA cores, thereby freeing the GPU's highly specialized Tensor Cores entirely for deep learning inference tasks.[41]

Within the DeepStream pipeline, incoming streams are aggregated and batched together via the gst-nvstreammux plugin.[43] If the batch size is configured to 32, a single GPU can infer across 32 distinct polling stations simultaneously. The memory allocation within gst-nvstreammux is dynamically managed; ensuring that the input stream resolutions match the muxer output resolution is critical to preventing exorbitant GPU memory overhead caused by real-time scaling operations.[43]

3.3 Evaluating Computer Vision Models for Low-Resolution, Angled Cameras

The selection of the primary object detection model involves navigating a critical trade-off between inference latency (speed) and Mean Average Precision (mAP). Recent benchmarks provide vital insights into model selection for surveillance environments:

- **YOLOv8:** Benchmarks indicate that YOLOv8 provides exceptional recall and a high Area Under the Curve (AUC-PR), effectively minimizing false negatives.[45] Its architecture, specifically the YOLOv8s (small) variant, is highly optimized for edge and real-time processing, achieving frame rates exceeding 170 FPS on modern GPUs while maintaining robust detection quality.[47] However, it may exhibit higher false positive rates in visually

cluttered scenes compared to newer models.[45]

- **YOLOv9 and YOLOv11:** These newer architectures incorporate advanced techniques such as Programmable Gradient Information (PGI), which optimizes gradient flow during training, significantly improving accuracy in highly complex, occluded environments.[49] While heavier variants like YOLOv11-x provide unmatched detection accuracy, they introduce higher latency that may bottleneck a high-density RTSP pipeline.[48]

For the specific context of the Armenian election, where cameras will likely be mounted at angled, suboptimal vantage points in cramped municipal rooms or schools, generating low-resolution and highly skewed data, the **YOLOv8s** or **YOLOv11-s** models are strongly recommended. They provide the necessary mathematical equilibrium: sufficiently lightweight to permit the massive batch sizes required to process 1800 streams, yet sophisticated enough to consistently detect persons, hands, and ballot boxes.[47] To maximize throughput, these models must be exported to TensorRT format, yielding up to a 5x GPU speedup compared to standard PyTorch execution.[51]

3.4 Multi-Object Tracking (MOT), ReID, and Overcoming Occlusions

Polling stations are structurally chaotic environments. Voters, commission members, domestic observers, and media personnel will constantly intermingle, frequently blocking the camera's field of view. The primary algorithmic challenge is tracking individuals across frames without losing their assigned ID when they cross behind another person or an object (occlusion). Standard trackers, such as DeepSORT, struggle profoundly in crowded scenarios because their reliance on bounding box Intersection-over-Union (IoU) fails when boxes overlap completely. To overcome this, **PD-SORT (Pseudo-Depth SORT)** is highly recommended. PD-SORT is specifically engineered to be occlusion-robust; it generates pseudo-depth cues to estimate the relative distance of objects from the camera, creating a 2.5D representation of the scene.[52] By leveraging depth information during the matching phase, PD-SORT effectively manages spatial overlaps, preventing the catastrophic ID-switching that plagues traditional trackers during dense crowding.[52] Furthermore, utilizing Kalman filters, PD-SORT maintains temporal persistence, predicting a person's trajectory even if they are completely occluded for up to 60 frames.[52]

For the ReID embedding extraction necessary to enforce the single-assistance rule of Article 65, the system should deploy **Sh-ViT (Shuffling Vision Transformer)**. Sh-ViT is a recent, lightweight model explicitly engineered for occluded person ReID in complex surveillance environments.[28] By shuffling patches in the final Transformer layers, it deliberately breaks spatial correlations, forcing the model to learn robust, localized features rather than relying on the overall shape of the person.[28] This makes Sh-ViT highly resilient to partial occlusions, motion blur, and poor camera angles—the exact hostile conditions anticipated in Armenian polling stations.[28] Because it evaluates pedestrians based on local visual tokens, it can achieve positive identity matches even when a subject is only partially visible (e.g., matching the pattern of a shirt when the lower body is obscured by a desk).[28]

3.5 Cloud Infrastructure Requirements, Cost Estimation, and Latency

Bottlenecks

Executing the proposed multi-stage pipeline (H.265 Decode → nvstreammux → TensorRT YOLOv8s → PD-SORT → Sh-ViT ReID) on 1800 concurrent streams demands formidable compute power. Profiling data for NVIDIA A100 GPUs indicates that a single A100 (80GB) can sustain approximately 60 to 80 simultaneous 720p streams, heavily dependent on the complexity and batching frequency of the ReID module.[55]

To process 1800 streams reliably, incorporating a generous safety buffer for unexpected load spikes, the architecture requires approximately 30 to 40 NVIDIA A100 GPUs. Leveraging a hyperscale cloud provider such as Amazon Web Services (AWS) is the most viable path, utilizing their specialized **Amazon EC2 P4d Instances**.

- A single p4d.24xlarge instance houses 8x A100 Tensor Core GPUs, alongside 96 vCPUs and 1152 GiB of memory, backed by 400 Gbps networking.[59]
- Five (5) p4d.24xlarge instances would provide the requisite 40 A100 GPUs.
- The on-demand pricing for a p4d.24xlarge instance in the US East region is approximately $32.77 per hour.[61]

Infrastructure Component	**Quantity**	**Hourly Rate (USD)**	**72-Hour Run Cost (Pre-election, Election Day, Recount)**
AWS p4d.24xlarge (8x A100)	5 Instances	$32.77 / instance	~$11,797.20
EC2 Network Data Egress/Ingress	~10 TB transfer	Variable	~$1,000.00
Load Balancers & Media Servers	10 c5.4xlarge	~$0.68 / instance	~$489.60
Total Cloud Compute Estimate			**~$13,286.80**

It is imperative to note that this estimate covers *only* the raw cloud compute expenditure during the immediate 72-hour operational window. It does not account for the extensive, multi-month software engineering required, the vast costs of physical camera procurement and installation by ISPs, or the custom model training. Historically, the Armenian government has allocated significant budgets for electoral organization—approving an $11.8 million cost estimate for the 2026 elections and spending $12.2 million in bonuses related to the 2021 elections [62]—indicating that the capital exists for large-scale infrastructure projects.

The primary operational bottleneck will not be AI inference latency, which operates in the realm of milliseconds per frame [48], but rather the fragility of network transmission from rural, geographically challenging regions (e.g., Syunik, Gegharkunik) to the centralized cloud infrastructure.[64] Packet loss over varying cellular or degraded fixed networks can cause stream corruption, resulting in dropped frames and pipeline crashes.[65] The DeepStream architecture

must be meticulously configured to handle intermittent connection drops gracefully, utilizing parameters such as batched-push-timeout to ensure the core inference engine does not stall indefinitely while waiting for a delayed packet from a rural stream.[65]

4. AI vs. Human Observers: Effectiveness, Cost-Benefit, and Synergy

In Armenia, the integrity of elections relies heavily on observation missions conducted by civil society organizations such as "Akanates" (Eyewitness), Transparency International Armenia, and the Union of Informed Citizens, alongside international bodies like the OSCE/ODIHR.[5] These groups deploy hundreds of short-term and long-term observers to monitor electoral processes, document abuses of administrative resources, and provide critical, independent assessments of the election's legitimacy.[6] The introduction of an automated AI system does not render these human observers obsolete; rather, it radically alters their operational deployment, amplifying their capabilities through technological synergy.

4.1 The Cost-Benefit Calculus: Scalability vs. Contextual Nuance

The comparative analysis between deploying 1800 human observers and activating a nationwide AI network reveals stark contrasts in financial modeling, operational endurance, and analytical capabilities.

Financial and Logistical Costs: Deploying a human observer to every single polling station requires immense logistical coordination and continuous funding. Observers require extensive legal training, transportation to remote regions, daily stipends, and secure communication equipment.[66] While the exact budget for an Akanates or Transparency International mission is highly variable based on donor funding, covering 1800 stations comprehensively represents a massive financial undertaking. Conversely, as calculated in Section 3.5, the raw inference compute cost for nationwide AI monitoring over the critical 72-hour period is remarkably low, sitting under $15,000 USD. While the upfront capital expenditure for AI software R&D and physical camera deployment is vastly higher—likely requiring multi-million dollar contracts with telecom providers [39]—the marginal cost of analyzing an additional polling station via AI approaches zero.

Qualitative Operational Effectiveness:

- **Endurance and Uniformity:** Human observers inevitably suffer from physical fatigue, momentary lapses in attention, and the potential for intimidation or coercion in historically hostile districts. An AI system operates continuously, applying uniform mathematical rigor for the entire 12-hour voting window and the critical, high-stress recount period post-20:00 [26], immune to fatigue or intimidation.
- **Scale of Memory:** A human observer cannot reliably memorize the faces, clothing patterns, and physical dimensions of 1,500 individual voters traversing a busy polling station, rendering them largely incapable of detecting if an individual re-enters the station hours later to illicitly assist a second voter. The Sh-ViT ReID model natively solves this limitation with absolute mathematical precision, maintaining a flawless memory of

every detected embedding.[28]

- **Contextual Reasoning and Semantic Understanding:** This is where AI exhibits its most profound weakness. AI is fundamentally devoid of human context. It may flag a crowd surging forward as a "security incident" or "unauthorized crowding," completely unable to recognize that the crowd is merely reacting benignly to a spilled box of pens or a sudden medical emergency.[23] Human observers possess the deep semantic understanding necessary to instantly differentiate malicious electoral fraud from benign, structural chaos.

4.2 Formulating an Operational Strategy: The Hybrid Legal Command Center

Because computer vision models, regardless of their sophistication, will inevitably generate false positives due to occlusions and contextual misinterpretations, directly utilizing algorithmic outputs to automatically invalidate a precinct's results is neither legally permissible under the Armenian Electoral Code nor practically viable. Instead, the optimal operational strategy treats the AI infrastructure as an asynchronous, nationwide threat-detection engine designed specifically to drive a human rapid-response protocol.

The Central Command Center Workflow:

1. **Automated Flagging:** The DeepStream computer vision pipeline operates silently until it detects a mathematical violation of the coded parameters (e.g., an unauthorized object intersects the ballot stack during the Article 68 recount, or a ReID embedding is matched assisting a second voter under Article 65).
2. **Evidentiary Clip Generation:** Upon triggering an alert, the system immediately extracts a high-definition, 15-second video clip of the incident (buffering 10 seconds pre-event and 5 seconds post-event). It appends immutable metadata to the clip, including the Precinct ID, exact timestamp, Violation Classification, and the algorithm's Confidence Score.
3. **Legal Triage and Verification:** This evidentiary package is pushed instantaneously to a centralized dashboard situated in a secure facility in Yerevan. This dashboard is manned continuously by a dedicated team of trained electoral lawyers and civil society experts (e.g., drawn from the ranks of Akanates or Transparency International).[66]
4. **Human Adjudication:** A legal expert reviews the 15-second clip. If their contextual understanding determines it is a false positive (e.g., a voter simply dropped their ID card near the ballot box), they discard the alert, and the system logs the feedback to improve future model weights. If they verify the violation as genuine, the event is immediately escalated.
5. **Rapid Response Dispatch:** The Central Command contacts a mobile, highly trained rapid-response legal team deployed regionally across Armenia's Marzes. This team is dispatched directly to the offending precinct to intervene physically, file an official complaint in the precinct's register, demand rectification from the Precinct Electoral Commission chairperson, and immediately escalate the matter to the Territorial Electoral Commission (TEC) and the Central Electoral Commission (CEC).[4]

This hybrid operational strategy maximizes the profound synergy between machine scalability and human legal expertise. The AI acts as an omnipresent, tireless filter, autonomously searching the equivalent of 21,600 hours of video (1800 stations × 12 hours) to surface the exact 15-second intervals that require human legal intervention. Furthermore, the generated video clips serve as irrefutable, timestamped evidentiary artifacts. When domestic observer missions or political parties submit formal complaints to the Central Electoral Commission or the courts, these clips significantly strengthen their legal standing, moving disputes away from hearsay and toward objective visual evidence.[5]

4.3 Realistic Assessment of Practical Feasibility for 2026

Evaluating whether this conceptual framework is practically feasible for a nation-wide rollout in the Republic of Armenia by the 2026 Parliamentary Elections yields a definitive conclusion: **It is feasible, but strictly contingent upon immediate development and rigorous sociopolitical safeguarding.**

From a purely technological standpoint, the hardware and algorithmic components are entirely commoditized and proven. The national telecommunications infrastructure, spearheaded by entities capable of massive deployments like Ucom and Team Telecom Armenia, has already proven capable of streaming 1500+ precincts simultaneously during the intense 2021 snap elections.[38] The requisite cloud infrastructure can be seamlessly and elastically leased on-demand from hyperscale providers like AWS. Open-source, cutting-edge AI architectures like YOLOv8/v11, PD-SORT, and Sh-ViT are freely available and can be rapidly containerized for deployment via NVIDIA DeepStream.[28]

However, the primary barriers to viability are not technical, but rather developmental, legal, and sociopolitical:

1. **Software Development Lead Time and Custom Training:** An AI architecture of this specialized complexity requires at least 12 to 18 months of dedicated software engineering, edge-case testing, and MLOps pipeline construction. Training the object detection and tracking models on custom datasets representing the specific lighting, spatial geometry, and demographic appearance of Armenian schools and municipal buildings is absolutely mandatory to prevent unacceptable false-positive rates.[70] Off-the-shelf models trained on Western urban environments will fail in rural Armenian precincts.
2. **Legal Compliance and Public Trust:** Implementing pervasive AI surveillance in a highly polarized society runs the severe risk of generating intense public skepticism and accusations of state overreach, potentially undermining the very democratic trust the system aims to protect.[4] The system must be radically transparent. The source code, the data anonymization protocols, and the absolute exclusion of facial recognition in favor of generalized, appearance-based ReID must be audited by independent third parties and international bodies like the OSCE/ODIHR. This is the only way to ensure total compliance with the Law on Protection of Personal Data and maintain the fragile trust of the electorate.[30]

If these developmental and legal mandates are met, the proposed system possesses the

unprecedented potential to systematically elevate the integrity, transparency, and defensibility of Armenia's democratic process.

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